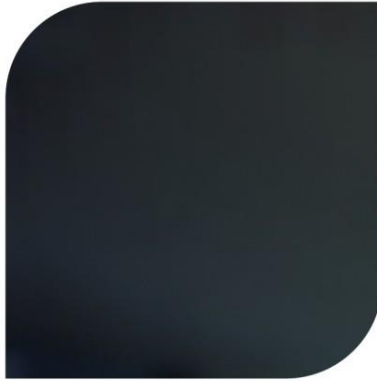
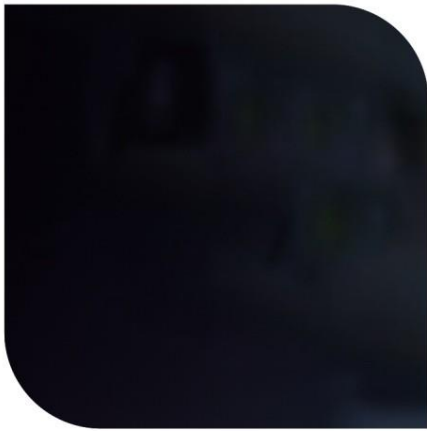


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Smart PG Management System Project Report

A report Submitted in fulfillment of the requirements for the organization



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Abstract

The Smart PG Hostel Management System is a console-based application developed using Python to simplify and automate hostel and paying guest management activities. The system is designed to help hostel administrators efficiently manage resident records, room allocation, rent payments, visitor entries, and room status details in a centralized manner. The project reduces manual paperwork and improves the accuracy and speed of hostel operations. It includes a secure admin login system where only authorized users can access the management features. The application stores resident information such as resident name, age, phone number, ID proof, sharing type, room allocation, rent details, and joining date. The project also validates mobile numbers by accepting only 10-digit phone numbers, ensuring proper data entry. Different sharing options like 2-sharing, 3-sharing, and 4-sharing rooms are included with separate rent structures. The rent payment module allows the administrator to track paid and pending rents easily. Additional features such as visitor entry records, room occupancy status, resident search functionality, and hostel reports make the system more practical and user-friendly. The project demonstrates the implementation of Python concepts such as functions, dictionaries, lists, loops, conditional statements, and exception handling in a real-time application. Overall, the Smart PG Hostel Management System provides an efficient, organized, and reliable solution for managing hostel activities digitally.

Introduction

The rapid growth of students and working professionals moving to different cities for education and employment has increased the demand for paying guest hostels and accommodation facilities. Managing a hostel manually becomes difficult when the number of residents increases. Traditional hostel management methods involve maintaining records in registers and files, which consumes more time and may lead to errors, data loss, and inefficient management. To overcome these problems, the Smart PG Hostel Management System is developed as a console-based Python application that helps automate hostel management activities efficiently.

The Smart PG Hostel Management System is designed to manage all important hostel operations in a simple and organized way. The system provides secure admin authentication to ensure that only authorized persons can access and manage the hostel data. After successful login, the administrator can perform various operations such as adding residents, viewing resident details, allocating rooms, managing rent payments, recording visitor entries, viewing room status, searching

resident information, and generating hostel reports. The application maintains all records digitally, reducing manual effort and improving accuracy.

One of the important features of the system is resident management. The administrator can add new residents by entering details such as resident name, age, phone number, ID proof, and room sharing type. The system validates the phone number and accepts only valid 10-digit mobile numbers to maintain proper data consistency. Different room-sharing options are provided, including 2-sharing, 3-sharing, and 4-sharing rooms, each having different rent amounts. The system checks whether rooms are vacant or occupied before allocation, preventing duplicate room assignments. The rent payment module allows administrators to track paid and unpaid rent status for each resident. The administrator can easily identify pending rents and manage hostel income effectively. The visitor entry feature maintains records of visitors entering the hostel, including visitor name, resident name, and entry time, which improves hostel security and monitoring.

The room status module displays the occupancy details of all rooms, showing whether a room is vacant or occupied. The resident search feature helps quickly find resident information without checking all records manually. The hostel report module provides an overview of total residents, rent payment status, pending payments, and total visitors, helping the administrator analyze hostel operations efficiently.

Concepts Used

- Functions
- Conditional Statements
- Loops
- Lists
- Dictionaries
- Exception Handling
- Date and Time Module
- Input Validation
- Nested Conditions
- Menu Driven Programming

Key Features

- Admin Login System
- Add Resident Details
- Mobile Number Validation
- Room Sharing Management
- Automatic Rent Calculation
- Room Allocation System
- Rent Payment Tracking
- Visitor Entry Management
- View Resident Records
- Search Resident Feature
- Room Status Monitoring
- Hostel Report Generation
- Occupied and Vacant Room Checking
- Console Based User Interface

Project Name: Smart PG Management System

Source Code

pg.py > add_resident

```
1 # =====
2 # SMART PG HOSTEL MANAGEMENT SYSTEM
3 # Console Based Python Project
4 # =====
5 import datetime
6 # ----- LOGIN DETAILS -----
7 ADMIN_USERNAME = "admin"
8 ADMIN_PASSWORD = "1234"
9 # ----- DATA STORAGE -----
10 residents = []
11 visitors = []
12 rooms = {
13     101: None,
14     102: None,
15     103: None,
16     104: None,
17     105: None
18 }
19 # ----- FUNCTIONS -----
20 def login():
21     print("\n===== ADMIN LOGIN =====")
22     username = input("Enter Username: ")
23     password = input("Enter Password: ")
24     if username == ADMIN_USERNAME and password == ADMIN_PASSWORD:
25         print("\nLogin Successful!")
26         return True
27     else:
28         print("\nInvalid Username or Password!")
29         return False
30 # -----
31 def add_resident():
32     print("\n===== ADD RESIDENT =====")
33     name = input("Enter Resident Name: ")
34     age = input("Enter Age: ")
35     phone = input("Enter Phone Number: ")
36     if not(phone.isdigit() and len(phone)==10):
37         print("Invalid Phone Number!")
38         return
39     id_proof = input("Enter ID Proof Number: ")
40     print("\nRoom Sharing Types")
41     print("2 Sharing --> 8000")
42     print("3 Sharing --> 7000")
43     print("4 Sharing --> 6000")
44     sharing = input("Enter Sharing Type: ")
45     if sharing == "2":
46         rent = 8000
47     elif sharing == "3":
```

```
48         rent = 7000
49     elif sharing == "4":
50         rent = 6000
51     else:
52         print("Invalid Sharing Type!")
53         return
54     resident = {
55         "name": name,
56         "age": age,
57         "phone": phone,
58         "id_proof": id_proof,
59         "sharing": sharing,
60         "rent": rent,
61         "room": None,
62         "rent_paid": False,
63         "join_date": str(datetime.date.today())
64     }
65     residents.append(resident)
66     print("\nResident Added Successfully!")
67 # -----
68 def view_residents():
69     print("\n===== RESIDENT LIST =====")
70     if len(residents) == 0:
71         print("No Residents Found!")
72         return
73     count = 1
74     for resident in residents:
75         print(f"\nResident {count}")
76         print("Name      :", resident["name"])
77         print("Age       :", resident["age"])
78         print("Phone     :", resident["phone"])
79         print("ID Proof  :", resident["id_proof"])
80         print("Sharing   :", resident["sharing"])
81         print("Rent      :", resident["rent"])
82         print("Join Date :", resident["join_date"])
83         if resident["room"] is None:
84             print("Room      : Not Allocated")
85         else:
86             print("Room      :", resident["room"])
87         if resident["rent_paid"]:
88             print("Rent Status: Paid")
89         else:
90             print("Rent Status: Pending")
91         count += 1
92 # -----
```

```

93 def allocate_room():
94     print("\n===== ROOM ALLOCATION =====")
95     if len(residents) == 0:
96         print("No Residents Available!")
97         return
98     name = input("Enter Resident Name: ")
99     resident_found = None
100    for resident in residents:
101        if resident["name"].lower() == name.lower():
102            resident_found = resident
103            break
104    if resident_found is None:
105        print("Resident Not Found!")
106        return
107    if resident_found["room"] is not None:
108        print("Room Already Allocated!")
109        return
110    print("\nAvailable Rooms:")
111    available = False
112    for room_no, occupant in rooms.items():
113        if occupant is None:
114            print("Room", room_no)
115            available = True
116    if not available:
117        print("No Rooms Available!")
118        return
119    try:
120        room_choice = int(input("Enter Room Number: "))
121    except:
122        print("Invalid Input!")
123        return
124    if room_choice not in rooms:
125        print("Invalid Room Number!")
126        return
127    if rooms[room_choice] is not None:
128        print("Room Already Occupied!")
129        return
130    rooms[room_choice] = resident_found["name"]
131    resident_found["room"] = room_choice
132    print("\nRoom Allocated Successfully!")
133    # -----
134    def pay_rent():
135        print("\n===== RENT PAYMENT =====")
136        if len(residents) == 0:
137            print("No Residents Available!")
138            return

```

```

139     name = input("Enter Resident Name: ")
140     for resident in residents:
141         if resident["name"].lower() == name.lower():
142             print("Amount To Pay :", resident["rent"])
143             if resident["rent_paid"]:
144                 print("Rent Already Paid!")
145             else:
146                 resident["rent_paid"] = True
147                 print("Rent Payment Successful!")
148             return
149     print("Resident Not Found!")
150     # -----
151     def visitor_entry():
152         print("\n===== VISITOR ENTRY =====")
153         visitor_name = input("Enter Visitor Name: ")
154         resident_name = input("Enter Resident Name To Visit: ")
155         found = False
156         for resident in residents:
157             if resident["name"].lower() == resident_name.lower():
158                 found = True
159                 break
160         if not found:
161             print("Resident Not Found!")
162             return
163         entry_time = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")
164         visitor = {
165             "visitor_name": visitor_name,
166             "resident_name": resident_name,
167             "entry_time": entry_time
168         }
169         visitors.append(visitor)
170         print("\nVisitor Entry Recorded Successfully!")
171         # -----
172         def view_visitors():
173             print("\n===== VISITOR RECORDS =====")
174             if len(visitors) == 0:
175                 print("No Visitor Records Found!")
176                 return
177             count = 1
178             for visitor in visitors:
179                 print(f"\nVisitor {count}")
180                 print("Visitor Name :", visitor["visitor_name"])
181                 print("Resident      :", visitor["resident_name"])
182                 print("Entry Time   :", visitor["entry_time"])
183                 count += 1
184             # -----

```

```

185 def room_status():
186     print("\n===== ROOM STATUS =====")
187     total_rooms = len(rooms)
188     occupied = 0
189     vacant = 0
190     for room_no, occupant in rooms.items():
191         if occupant is None:
192             status = "Vacant"
193             vacant += 1
194         else:
195             status = "Occupied"
196             occupied += 1
197         print(f"Room {room_no} --> {status}")
198     print("\nTotal Rooms   :", total_rooms)
199     print("Occupied Rooms:", occupied)
200     print("Vacant Rooms   :", vacant)
201 # -----
202 def search_resident():
203     print("\n===== SEARCH RESIDENT =====")
204     name = input("Enter Resident Name: ")
205     for resident in residents:
206         if resident["name"].lower() == name.lower():
207             print("\nResident Found!")
208             print("Name       :", resident["name"])
209             print("Age        :", resident["age"])
210             print("Phone      :", resident["phone"])
211             print("ID Proof   :", resident["id_proof"])
212             print("Sharing    :", resident["sharing"])
213             print("Rent       :", resident["rent"])
214             print("Join Date  :", resident["join_date"])
215             if resident["room"] is None:
216                 print("Room       : Not Allocated")
217             else:
218                 print("Room       :", resident["room"])
219             if resident["rent_paid"]:
220                 print("Rent Status: Paid")
221             else:
222                 print("Rent Status: Pending")
223             return
224     print("Resident Not Found!")
225 # -----
226 def hostel_report():
227     print("\n===== HOSTEL REPORT =====")
228     total_residents = len(residents)
229     paid = 0
230     pending = 0

```

```

231     for resident in residents:
232         if resident["rent_paid"]:
233             paid += 1
234         else:
235             pending += 1
236     print("Total Residents :", total_residents)
237     print("Rent Paid       :", paid)
238     print("Rent Pending    :", pending)
239     print("Total Visitors  :", len(visitors))
240 # -----
241 def main_menu():
242     while True:
243         print("\n=====")
244         print(" SMART PG HOSTEL MANAGEMENT SYSTEM ")
245         print("=====")
246         print("1. Add Resident")
247         print("2. View Residents")
248         print("3. Allocate Room")
249         print("4. Pay Rent")
250         print("5. Visitor Entry")
251         print("6. View Visitors")
252         print("7. Room Status")
253         print("8. Search Resident")
254         print("9. Hostel Report")
255         print("10. Exit")
256         choice = input("\nEnter Your Choice: ")
257         if choice == "1":
258             add_resident()
259         elif choice == "2":
260             view_residents()
261         elif choice == "3":
262             allocate_room()
263         elif choice == "4":
264             pay_rent()
265         elif choice == "5":
266             visitor_entry()
267         elif choice == "6":
268             view_visitors()
269         elif choice == "7":
270             room_status()
271         elif choice == "8":
272             search_resident()
273         elif choice == "9":
274             hostel_report()

```



```

275         elif choice == "10":
276             print("\nThank You For Using The System!")
277             break
278         else:
279             print("\nInvalid Choice! Please Try Again.")
280     # ----- MAIN PROGRAM -----
281     print("=====")
282     print("      WELCOME TO PG HOSTEL APP      ")
283     print("=====")
284     if login():
285         main_menu()
286     else:
287         print("\nAccess Denied!")

```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\python\mini projects\rohith> python3 pg.py
=====
          WELCOME TO PG HOSTEL APP
=====

===== ADMIN LOGIN =====
Enter Username: admin
Enter Password: 1234

Login Successful!

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====

1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 1

===== ADD RESIDENT =====
Enter Resident Name: rohith
Enter Age: 24
Enter Phone Number: 9515536330
Enter ID Proof Number: 1234567891

Room Sharing Types
2 Sharing --> 8000
3 Sharing --> 7000
4 Sharing --> 6000
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter Sharing Type: 2

Resident Added Successfully!

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====

1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 2

===== RESIDENT LIST =====

Resident 1
Name      : rohith
Age       : 24
Phone     : 9515536330
ID Proof  : 1234567891
Sharing   : 2
Rent      : 8000
Join Date : 2026-05-19
Room      : Not Allocated
Rent Status: Pending
```

```
=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit
```

Enter Your Choice: 3

```
===== ROOM ALLOCATION =====
Enter Resident Name: 101
Resident Not Found!
```

```
=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit
```

Enter Your Choice: 4

```
===== RENT PAYMENT =====
Enter Resident Name: rohith
Amount To Pay : 8000
Rent Payment Successful!
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 5

===== VISITOR ENTRY =====
Enter Visitor Name: nithin
Enter Resident Name To Visit: rohith

Visitor Entry Recorded Successfully!

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 6

===== VISITOR RECORDS =====

Visitor 1
Visitor Name : nithin
Resident    : rohith
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Entry Time   : 2026-05-19 12:07:41

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 7

===== ROOM STATUS =====
Room 101 --> Vacant
Room 102 --> Vacant
Room 103 --> Vacant
Room 104 --> Vacant
Room 105 --> Vacant

Total Rooms   : 5
Occupied Rooms: 0
Vacant Rooms  : 5

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 3

===== ROOM ALLOCATION =====
Enter Resident Name: rohith

Available Rooms:
Room 101
Room 102
Room 103
Room 104
Room 105
Enter Room Number: 101

Room Allocated Successfully!

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 7

===== ROOM STATUS =====
Room 101 --> Occupied
Room 102 --> Vacant
Room 103 --> Vacant
Room 104 --> Vacant
Room 105 --> Vacant

Total Rooms : 5
Occupied Rooms: 1
Vacant Rooms : 4

=====
SMART PG HOSTEL MANAGEMENT SYSTEM
=====
1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter Your Choice: 8

===== SEARCH RESIDENT =====

Enter Resident Name: rohith

Resident Found!

Name : rohith
Age : 24
Phone : 9515536330
ID Proof : 1234567891
Sharing : 2
Rent : 8000
Join Date : 2026-05-19
Room : 101
Rent Status: Paid

=====

SMART PG HOSTEL MANAGEMENT SYSTEM

=====

1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 9

===== HOSTEL REPORT =====

Total Residents : 1
Rent Paid : 1
Rent Pending : 0
Total Visitors : 1

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====

SMART PG HOSTEL MANAGEMENT SYSTEM

=====

1. Add Resident
2. View Residents
3. Allocate Room
4. Pay Rent
5. Visitor Entry
6. View Visitors
7. Room Status
8. Search Resident
9. Hostel Report
10. Exit

Enter Your Choice: 10

Thank You For Using The System!

PS D:\python\mini projects\rohith> █

Conclusion

The Smart PG Hostel Management System is an efficient and user-friendly console-based application developed using Python to simplify hostel and paying guest management activities. The project successfully manages important operations such as resident registration, room allocation, rent payment tracking, visitor management, room status checking, and hostel report generation. It reduces manual paperwork and helps administrators maintain records in an organized and systematic manner.

The system also provides useful features such as secure admin login, mobile number validation, room sharing options with different rent structures, and resident search functionality. These features make the application practical and suitable for real-world hostel management requirements. By using Python concepts like functions, loops, conditional statements, lists, dictionaries, and exception handling, the project demonstrates the implementation of programming fundamentals in a real-time application.

Overall, the Smart PG Hostel Management System improves accuracy, saves time, enhances data management, and provides a better approach for handling hostel operations digitally. The project serves as a valuable learning experience for understanding software development and management systems using Python programming.



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